



KARNATAKA SSLC PATTERN - SCIENCE PRACTICE QUESTION PAPER

S.S.L.C. / CLASS X - SCIENCE
English Medium | Subject Code: 83-E

Time: 3 Hours 15 Minutes	Max. Marks: 80
Portion: Full Science Syllabus	Total Questions: 38

General Instructions to the Candidate

1. There are three parts: **Part A - Physics, Part B - Chemistry, Part C - Biology.**
2. The paper contains **38 questions**. Follow the instructions given against each section.
3. Figures in the right-hand margin indicate the maximum marks allotted.
4. The maximum time includes **15 minutes for reading** the question paper.
5. Answer all questions unless an **OR** choice is provided.
6. Use SI units wherever applicable and show necessary steps in numerical problems.

This is an original practice paper prepared for learning purposes. It follows the Karnataka SSLC Science syllabus and recent KSEAB model-paper structure; it is not an official KSEAB paper.

PART - A

(PHYSICS)

I. Choose the correct alternative. **2 x 1 = 2**

1. The SI unit of electric power is

(A) volt (V) **(B)** watt (W) **(C)** ampere (A) **(D)** ohm

2. When an object is placed beyond 2F in front of a convex lens, the image is

(A) virtual and enlarged **(B)** real and diminished **(C)** virtual and diminished **(D)** real and enlarged

II. Answer the following questions. **2 x 1 = 2**

3. Write the symbols for a fuse and a switch used in an electric circuit.

4. If the current appears anticlockwise when viewed from one face of a circular coil, which magnetic pole is formed on that face?

III. Answer the following questions. **3 x 2 = 6**

5. State the two laws of reflection of light.

6. Mention any two factors on which the resistance of a conductor depends.

7. What is a solenoid? Mention one use of an electromagnet made using a solenoid.

IV. Answer the following questions. **3 x 3 = 9**

8. Draw the ray diagram for an object placed between F and 2F of a convex lens. State the position, size and nature of the image.

9. A 6 ohm resistor is connected to a 12 V battery. Find the current. If the current through the same resistor is 3 A, find the potential difference.

10. Explain the principle of an electric motor. Name the rule used to determine the direction of force on the conductor.

V. Answer the following questions. **2 x 4 = 8**

11. a) State Ohm's law. b) A 4 ohm and a 6 ohm resistor are connected in series to a 20 V source. Find the equivalent resistance and current.

12. a) Define power of a lens and give its SI unit. b) An object is 30 cm from a convex lens of focal length 20 cm. Find the image distance using the lens formula and state the nature of the image.

PART - B
(CHEMISTRY)

VI. Choose the correct alternative. **3 x 1 = 3**

13. The gas evolved when dilute hydrochloric acid reacts with zinc is
(A) oxygen **(B)** hydrogen **(C)** carbon dioxide **(D)** chlorine

14. The property of carbon that enables it to form long chains is
(A) tetravalency **(B)** catenation **(C)** neutralisation **(D)** corrosion

15. A solution having pH 3 is
(A) strongly acidic **(B)** weakly acidic **(C)** neutral **(D)** basic

VII. Answer the following questions. **3 x 1 = 3**

16. What is a balanced chemical equation?

17. Why is sodium stored under kerosene?

18. What is rancidity? Mention one method of preventing it.

VIII. Answer the following questions. **3 x 2 = 6**

19. Write balanced equations for (i) magnesium reacting with oxygen and (ii) calcium carbonate undergoing thermal decomposition.

20. a) What is the pH scale used for? b) Give one acidic and one basic substance used in everyday life.

21. Explain the formation of sodium chloride by transfer of electrons between sodium and chlorine atoms.

IX. Answer the following questions. **3 x 3 = 9**

22. Write balanced equations and identify the type of reaction for: (i) iron with copper sulphate solution, (ii) silver chloride exposed to sunlight, and (iii) an acid reacting with a base.

23. a) What are ores? b) Name the main steps involved in obtaining a metal from its ore. c) Mention one method of preventing corrosion.

24. a) Define homologous series. b) Write the molecular formulae of the first two alkanes. c) Give one difference between saturated and unsaturated carbon compounds.

X. Answer the following question. **1 x 4 = 4**

25. a) Explain why carbon generally forms covalent compounds. b) Write the structural formulae of ethene and ethanoic acid. **OR** a) Explain the cleansing action of soap. b) What is esterification? Give one example.

PART - C

(BIOLOGY)

XI. Choose the correct alternative. **3 x 1 = 3**

26. The functional unit of the kidney is

(A) neuron **(B)** nephron **(C)** alveolus **(D)** villus

27. The plant hormone mainly associated with phototropism is

(A) auxin **(B)** insulin **(C)** adrenaline **(D)** thyroxine

28. Transmission of a nerve impulse from one neuron to another occurs across the

(A) alveolus **(B)** synapse **(C)** nephron **(D)** stoma

XII. Answer the following questions. **3 x 1 = 3**

29. What is double circulation?

30. What is DNA replication?

31. Why is variation important for the survival of a species?

XIII. Answer the following questions. **2 x 2 = 4**

32. a) Define reflex action. b) Write the sequence of events in a reflex arc.

33. a) What is vegetative propagation? b) Mention two advantages of vegetative propagation.

XIV. Answer the following questions. **3 x 3 = 9**

34. Draw a neat labelled diagram of the human heart. Label the right atrium, left ventricle and aorta.

35. A tall pea plant with genotype Tt is crossed with another Tt plant. Show the F₂ generation using a checkerboard and state the genotypic and phenotypic ratios.

36. a) What are biodegradable substances? b) What is biomagnification? c) Mention one cause of ozone depletion and one way to reduce environmental pollution.

XV. Answer the following question. **1 x 4 = 4**

37. a) State the functions of xylem and phloem. b) Explain briefly how gaseous exchange occurs in the alveoli of human lungs.

XVI. Answer the following question. **1 x 5 = 5**

38. a) Explain how a fertilised egg develops into an embryo in the uterus. b) Mention the role of the placenta. c) State any two methods of contraception. **OR** a) Explain sexual reproduction in flowering plants from pollination to fertilisation. b) State any two changes that occur in a flower after fertilisation.

SYLLABUS COVERAGE & PAPER BLUEPRINT

The paper is built around the Karnataka SSLC Class 10 Science syllabus and the 80-mark theory-paper structure reflected in recent KSEAB model papers. The full-syllabus coverage includes the following units:

Area	Units covered
Physics	Light - Reflection and Refraction; The Human Eye and the Colourful World; Electricity; Magnetic Effects of Electric Current; Sources of Energy
Chemistry	Chemical Reactions and Equations; Acids, Bases and Salts; Metals and Non-metals; Carbon and its Compounds; Periodic Classification of Elements
Biology	Life Processes; Control and Coordination; How do Organisms Reproduce?; Heredity and Evolution; Our Environment; Sustainable Management of Natural Resources

Paper structure: Physics - 27 marks; Chemistry - 25 marks; Biology - 28 marks; Total - 80 marks.

END OF QUESTION PAPER